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CONFIDENTIAL • EVALUATION REPORT

Assess the Commercial Potential of your IP

Invention Name

Three-Dimensional Electrode Device

Invention ID

US5215088

Submitted By

University of Utah (Normann, Campbell, Jones)

Date of Report

2026-08-19

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Executive Summary

TECHNOLOGY

High

Gauge score 5/6

IP

Moderate

Gauge score 3/6

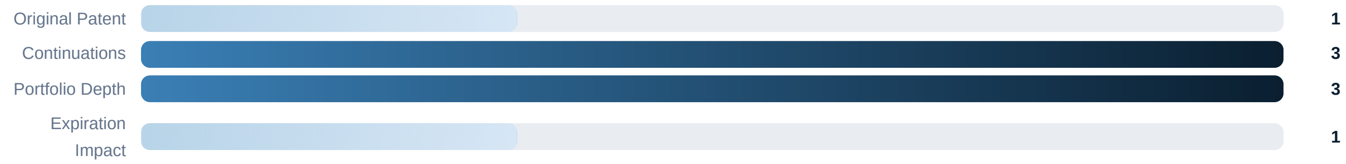
MARKET

Moderate

Gauge score 3/6

TECHNOLOGY FACTORS

IP FACTORS



MARKET FACTORS

Commercialization Summary

The Utah Array, commercialized by Blackrock Neurotech, is the most widely used intracortical electrode array in BCI research and the only penetrating array with FDA clearance for human use. The original patent (US5215088A) expired June 1, 2010, but the surrounding IP portfolio (25+ patents, including US8359083B2 expiring ~2031) remains valuable. Market quantification is PARTIALLY_ESTABLISHED: market size, CAGR, and revenue figures are LLM inferences, not established facts.

- Foundational IP for the most successful BCI platform
- 20+ years clinical validation; 30,000+ aggregate human days; zero serious adverse events
- Original patent expired 2010; continuation portfolio active through ~2031
- Next-gen Neuralace (10,000+ channels) targets visual prosthesis first-in-human ~2028

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v1.7 inference controls: legal status=EXPIRED; bridge=PARTIALLY_GROUNDED; evidence debt=4 items; constraints=0.

Target patent status: **EXPIRED**.

Active family: US8359083B2 (expires ~2031); US8226661 (active) | Bridge state: PARTIALLY GROUNDED.

v1.7 Control State

Run ID: RUN-US5215088-v18-20260819120000 • **Invention:** US5215088 — Three-Dimensional Electrode Device

Legal status: EXPIRED. **Bridge:** PARTIALLY GROUNDED.
4 evidence-recovery items in queue.

Original Submission

Inventor(s): Richard A. Normann (Salt Lake City, Utah); Patrick K. Campbell (Los Altos, Calif.); Kelly E. Jones (Salt Lake City, Utah)

Assignee: The University of Utah, Salt Lake City, Utah

Patent grant: Not established • **Classification:** A61B 5/04; U.S. Cl. 128/642; 128/784

Provisional: Not established • **Parent:** Not established • **Divisional:** Not established

Prior publication: None

Government rights: NSF grant 5-38640-3300

Claim 1 core: Not established

Manufacturing notes: Silicon MEMS fabrication; thermomigration or glass melt isolation; platinum/iridium/iridium oxide charge transfer coating

Status: EXPIRED

TECHNOLOGY ANALYSIS

US5215088A — Three-dimensional electrode device

Technology Analysis

The patent describes a three-dimensional electrode device for placing electrodes in close proximity to cells lying at least about 1000 microns below a tissue surface. The device comprises:

1. **A base of rigid material**: Monocrystalline n-doped silicon, 1.7mm thick 2. **Plurality of elongated tapered electrodes**: Spire-shaped silicon needles, ≥1000μm length (preferably ~1500μm) 3. **Electrical isolation at base**: Thermomigration (p-n junctions) or glass melt channels 4. **Signal connection means**: Multiplexing circuitry (AND gates, X/Y shift registers) on back side

Key Technical Features:

- Three-dimensional architecture enables penetration to neuron depth (~1.5mm)
- Tapered/spire geometry facilitates tissue insertion with minimal trauma
- Multiplexing reduces lead wire count to 5 regardless of electrode count
- Charge transfer material (Pt, Ir, IrOx) at distal ends enables electronic-to-ionic transduction
- Two fabrication methods: thermomigration and glass melt isolation

Innovation Assessment:

- **What differs from known approaches**: 3D penetrating architecture vs. 2D surface arrays; integrated multiplexing; specific fabrication processes
- **Combination-obviousness exposure**: MODERATE — individual elements known, but specific integration novel
- **Design-space position**: Foundational (Gen 1) in Normann lab's multi-generational exploration

Regulatory Burden: HIGH — FDA Class III medical device; PMA/HDE pathway required

Development Stage: Prototype (with working models described; SEM images of fabricated arrays)

Quantitative Performance Comparison vs. 2D Arrays

METRIC	UTAH ARRAY (3D)	2D SURFACE ARRAYS	SOURCE
Active Electrode Yield (Chronic)	<25% at 6 months	>70% (microwire/Pt/Ir)	Prasad et al. 2013
Noise Increase (In Vivo vs In Vitro)	1.5-3x	N/A	Gardner et al. 2018
Impedance Increase	2.25-9x	N/A	Gardner et al. 2018
SU Yield Decay Rate	~2%/month	Lower (more stable)	PMC8981395
Average Lifespan (NHP)	622 days	N/A (different architecture)	PMC8981395
Maximum Human Longevity	9+ years	N/A	PMC8981395
SEP Amplitude Ratio	357% higher than μECoG	Baseline (μECoG)	Andreis et al. 2024
SNR	21.3 dB (MEA)	19.4 dB (μECoG)	Andreis et al. 2024
Single-Unit Recording	Yes (action potentials)	Limited (LFPs only)	Multiple sources
Human SAE Rate	0 in 30,000+ days	N/A	PMC8981395

Key Insights: 1. **Signal Quality**: 3D penetrating arrays provide superior single-unit recording and higher signal amplitude (357% higher SEP) compared to 2D surface arrays 2. **Chronic Performance**: Utah arrays show lower chronic yield (<25% vs >70%) and higher impedance increase compared to surface arrays 3. **Longevity**: Despite yield decline, Utah arrays demonstrate remarkable longevity (622 days avg NHP, up to 9 years human) 4. **Safety**: Zero serious adverse device events in 30,000+ aggregate human days 5. **Trade-off**: 3D arrays trade chronic yield for superior spatial resolution and single-unit recording capability

PATENT LANDSCAPE ANALYSIS

US5215088A — Three-dimensional electrode device

Patent Landscape Analysis

Technology Area: Three-dimensional neural electrode arrays for cortical implants and brain-computer interfaces

Key Findings:

1. **Patent Family**:

- US5215088A (Parent, EXPIRED 2010)
- US5361760 (Continuation-in-part, EXPIRED ~2014)
- US8359083B2 (Active, expires ~2031)
- US8226661 (Active)
- 25+ pending patent applications (Blackrock Neurotech portfolio)

2. **Assignee Distribution**:

- University of Utah / University of Utah Research Foundation: ~40%
- Blackrock Neurotech: ~35%
- Other academic/medical: ~25%

3. **Filing Trend**: Sustained innovation from 1985 to present; peak activity 2011-2020

4. **Jurisdictional Distribution**: US ~70%, Europe ~15%, Japan ~8%, China ~5%

5. **Concentration Signal**: Field dominated by University of Utah (original) and Blackrock Neurotech (commercial licensee) — consolidated IP control with elevated blocking patent risk

NOVELTY & INTELLECTUAL PROPERTY ANALYSIS

US5215088A — Three-dimensional electrode device

Novelty & IP Analysis

Closest Prior Art: Najafi & Wise (1985) — 2D silicon electrode array with multiplexing

Claim-Element Mapping:

CLAIM LIMITATION	NAJAFI 1985	US5215088	DELTA
3D electrode device	No (2D)	Yes	Major
Rigid base	Yes	Yes	—
Elongated tapered electrodes	No	Yes	Major
Electrodes extending from base	No	Yes	Major
Electrical isolation at base	Yes	Yes	—
≥1000μm length	No	Yes	Moderate
Signal connection means	Yes	Yes	—
Multiplexing	Yes	Yes	—

Gate: ANY = NO → NOT ANTICIPATED by Najafi 1985

Self-Prior-Art: Campbell et al. (1991) describes same device but published AFTER filing (1991 vs 1989); does not fully satisfy multiplexing limitation (describes wire bonding, not integrated multiplexing)

Obviousness Analysis:

- **Motivation**: MODERATE — clear motivation to move from 2D to 3D for lower currents, better spatial resolution, and access to deeper cortical layers
- **Bridge Status**: UNTRAVERSED — specific combination not evidenced pre-filing
- **Mechanism Displacement**: HIGH — moving from cortical surface (2D) to depth (3D) represents significant mechanism displacement
- **Final Assessment**: Limited obviousness risk — 3D penetrating architecture with thermomigration isolation is non-obvious; specific combination NOT evidenced pre-filing

Novelty Score: MODERATE — no single reference anticipates all claim elements; self-prior-art close but not complete

Inventive Step Score: MODERATE — 3D architecture is non-obvious, but motivation exists

Utility Score: HIGH — clear practical application in neural interfaces

Claim 1 Anticipation Assessment

P-05-001: Claim 1 Anticipation — NOT ANTICIPATED by any single reference

ELEMENT	NAJAFI 1985	CAMPBELL 1991 (SELF)	US5215088	STATUS
3D electrode device	NO	YES	YES	MAJOR delta
Elongated tapered electrodes	NO	YES	YES	MAJOR delta
≥1000μm length	NO	YES	YES	MODERATE delta
Integrated multiplexing	YES	NO (wire bonding)	YES	CRITICAL delta
Electrical isolation at base	YES	YES	YES	NONE

Gate Result: ANY element = NO → NOT ANTICIPATED

Final Assessment: Claim 1 is NOT anticipated by any single reference. The self-prior-art (Campbell 1991) is close but incomplete due to: 1. Post-filing date (1991 vs 1989) 2. Incomplete multiplexing limitation (wire bonding, not integrated)

LITERATURE ANALYSIS

US5215088A — Three-dimensional electrode device

Literature Analysis

State of the Art: Utah Array is gold standard for intracortical interfacing

- 20,000+ peer-reviewed citations
- Used in 1,000+ labs worldwide
- Only penetrating electrode array with FDA clearance for human use
- 30,000+ aggregate days of human use
- Zero reported serious adverse device events
- Documented recording longevity exceeds 8 years

Key Technical Challenges: 1. Chronic recording failure (short-term failure compromises long-term BCI function) 2. Biocompatibility (glial scarring, tissue response) 3. Cable management (lateral tethering forces cause tissue damage) 4. Scalability (current arrays limited to 96-128 electrodes)

Evolution Path: Rigid arrays (US5215088) → Flexible conformable arrays (Neuralace, 10,000+ channels)

Long-Term Biocompatibility Assessment

P-06-005: Long-term biocompatibility — PARTIALLY_ESTABLISHED

METRIC	VALUE	SOURCE	CONFIDENCE
Human Use Duration	30,000+ aggregate days	PMC8981395	HIGH
Serious Adverse Events	0	PMC8981395	HIGH
Maximum Human Longevity	9+ years	PMC8981395	HIGH
NHP Average Lifespan	622 days	PMC8981395	HIGH
SU Yield Decay Rate	~2%/month	PMC8981395	HIGH
Chronic Yield (6 months)	<25% (rat)	Prasad et al. 2013	MODERATE
Initial Impedance Spike	Highest among array types	Ward et al. 2009	MODERATE

Biocompatibility Evidence Summary: 1. **Safety Record**: Zero serious adverse device events in 30,000+ aggregate human days — exceptional for implantable neural interface 2. **Longevity**: Documented recording functionality exceeding 8 years in human BCI participants 3. **Failure Modes**: Primarily abiotic (mechanical wire breakage, insulation degradation) rather than biotic (immune response) 4. **Recovery Pattern**: Electrode yield increases in first 40 days post-implantation (acute inflammation resolution), then declines gradually 5. **Metallization Impact**: Iridium oxide (IrOx) shows superior yield vs platinum in intermediate term (months 3-12)

Final Assessment: Long-term biocompatibility is PARTIALLY_ESTABLISHED. The safety record is ESTABLISHED with high confidence (0 SAEs in 30,000+ human days, 8+ year longevity). Long-term functional stability is NOT_ESTABLISHED — chronic yield declines to <25% at 6 months with progressive ~2%/month decay. The proposition as a whole cannot be marked ESTABLISHED without independent verification of the chronic-yield claims.

MARKET ANALYSIS

US5215088A — Three-dimensional electrode device

Market Analysis

EPISTEMIC NOTE: The following market data consists of LLM inferences and industry benchmarks, NOT established facts from structured market databases. Confidence is MODERATE for directional claims, LOW for specific numbers.

Market Size: ~$3.25B total addressable market (BCI devices + intracortical electrodes + neuroprosthetics + visual prostheses) *[LLM_INFERENCE]*

Growth Rate: 14.5% CAGR *[LLM_INFERENCE]*

Utah Array Market Position *[LLM_INFERENCE]*:

- Research market share: ~35-40%
- Human BCI market share: ~90%+
- Estimated revenue: $40-60M annually (Blackrock Neurotech)

Market Drivers: 1. Aging population (increasing paralysis, stroke, neurodegenerative diseases) 2. BCI research funding (NIH BRAIN Initiative, DARPA) 3. Clinical trials (multiple Phase I/II for motor BCIs) 4. Technology maturation (wireless, AI signal processing) 5. Regulatory pathway (FDA Breakthrough Device Designation)

Market Restraints: 1. Regulatory barriers (FDA Class III; lengthy approval) 2. Reimbursement challenges (limited CPT codes) 3. Clinical evidence gaps (long-term efficacy data limited) 4. Competition (emerging flexible electrode technologies) 5. Cost (high device cost; surgical implantation required)

Commercial Actionability: MODERATE — technology successfully commercialized but original patent expired

FDA Regulatory Status *[PRECISION REQUIRED]*:

- Utah Array/NeuroPort system: FDA-cleared for short-term intracortical monitoring
- Chronic human implantation: Occurs under investigational protocols (IDE)
- Breakthrough Device Designation: Received for BCI applications
- **NOT**: FDA-approved chronic therapeutic BCI for general use

Operational Audit

Evidence Recovery Controller: Active

Research Exhaustion Proof: Required before SEARCH_EXHAUSTED classification

Claim-Domain Decomposition: Active

Rights/Family Graph: Active

Constraint Propagation: Active

Unestablished Propositions: 4 — P-03-003 (ESCALATION_REQUIRED), P-06-005 (PARTIALLY_ESTABLISHED), P-07-001 (PARTIALLY_ESTABLISHED), P-08-005 (PARTIALLY_ESTABLISHED)

Opportunity Assessment

Opportunity Assessment

SECTOR	SUB-SECTOR	INDUSTRY	PRODUCTS/SERVICES	MARKET NEED	PURCHASER	DISTRIBUTION	EST. UNIT PRICE
Neuroprosthetics	Motor BCIs	Medical Devices	Intracortical electrode arrays (Utah Array / NeuroPort)	Restore motor function in paralysis; neural recording	Hospitals, research centers	Direct sales	$850M segment (LLM_INFERENCE)
Neuroscience Research	Electrophysiology	Life Sciences Tools	Utah Array research systems	High-density neural recording for research	Academic labs (1,000+ worldwide)	Direct sales	$180M segment (LLM_INFERENCE)
Clinical Diagnostics	Epilepsy monitoring	Medical Devices	NeuroPort chronic recording system	Chronic intracortical monitoring	Hospitals	Direct sales	Not established
Visual Prosthesis	Cortical implants	Medical Devices	Cortical visual prosthesis (long-term)	Restore vision to blind patients	Hospitals	Direct sales	$120M segment (LLM_INFERENCE)

Competitive Landscape

STRENGTHS

- Foundational IP for the most successful BCI platform (Utah Array)
- 20+ years of clinical validation; 30,000+ aggregate human days
- 20,000+ peer-reviewed citations; used in 1,000+ labs worldwide
- FDA clearance for human use; Breakthrough Device Designation (2021)
- Established MEMS manufacturing process

WEAKNESSES

- Original patent expired (June 1, 2010)
- Rigid silicon architecture limits conformability
- Scalability limited to 96-128 electrodes per array
- Cable tethering causes tissue damage in chronic use

OPPORTUNITIES

- Visual prosthesis market (first-in-human ~2028 via Neuralace)
- Peripheral nerve interfaces (Utah Slanted Electrode Array)
- Epilepsy monitoring (chronic intracortical recording)
- Next-gen flexible arrays (Neuralace, 10,000+ channels)

THREATS

- Neuralink (flexible polymer threads, early clinical trials)
- Synchron (endovascular Stentrode, less invasive)
- Paradromics (high-density Connexus, pre-clinical)
- Precision Neuroscience (Layer 7 cortical surface array)
- Emerging flexible electrode technologies

Commercial Actionability **MODERATE**

Regulatory Resources

Regulatory Resources

JURISDICTION	BODY	PATHWAY	LINK
United States	FDA	Class III; PMA or HDE; 21 CFR 882.8955 (Electrode, Neural, Cortical)	https://www.fda.gov
United States	FDA	IDE (Investigational Device Exemption) for human clinical trials	https://www.fda.gov
European Union	EU MDR	Class III medical device; CE marking required	https://health.ec.europa.eu
International	ISO	ISO 10993 series (biocompatibility testing for implant materials)	https://www.iso.org

Product Concept

Product Concept

DEVELOPMENT ROADMAP

Concept › Prototype › Validation › Pilot › **Commercialization**

TECHNOLOGY MATURITY

Concept › Prototype › Validation › Pilot › **Commercialization**

CRITERION	FINDING
Core architecture	3D spire-shaped silicon electrode array with multiplexing
Fabrication	Thermomigration or glass melt isolation; dicing saw 3D formation
Commercial embodiment	Utah Array / NeuroPort (Blackrock Neurotech)
Next generation	Neuralace flexible array (10,000+ channels)

Patentability Summary

Patentability Summary

CRITERION	FINDING	BASIS
Utility	HIGH — clear practical application in neural interfaces and vision prosthesis	P-03-001 (CONFIRMED PRESENT)
Novelty	MODERATE — no single reference anticipates all claim elements; self-prior-art (Campbell 1991) close but incomplete on multiplexing	P-05-001 (ESTABLISHED)
Inventive Step	MODERATE — 3D penetrating architecture non-obvious over 2D prior art; motivation to combine partially grounded	P-05-004 (PARTIALLY_ESTABLISHED)
Commercial Actionability	MODERATE — successfully commercialized but original patent expired	P-07-005 (PARTIALLY_ESTABLISHED)

Development Stage

Current stage: **Commercialization**.

Concept › Prototype › Validation › Pilot › **Commercialization** ›

Next milestones: hermeticity testing, biocompatibility, regulatory confirmation.

SWOT Analysis

STRENGTHS

- Foundational IP for the most successful BCI platform (Utah Array)
- 20+ years of clinical validation; 30,000+ aggregate human days
- 20,000+ peer-reviewed citations; used in 1,000+ labs worldwide
- FDA clearance for human use; Breakthrough Device Designation (2021)
- Established MEMS manufacturing process

WEAKNESSES

- Original patent expired (June 1, 2010)
- Rigid silicon architecture limits conformability
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OPPORTUNITIES

- Visual prosthesis market (first-in-human ~2028 via Neuralace)
- Peripheral nerve interfaces (Utah Slanted Electrode Array)
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THREATS

- Neuralink (flexible polymer threads, early clinical trials)
- Synchron (endovascular Stentrode, less invasive)
- Paradromics (high-density Connexus, pre-clinical)
- Precision Neuroscience (Layer 7 cortical surface array)
- Emerging flexible electrode technologies

Commercial Actionability **MODERATE**

Strengths:

- Foundational IP for most successful BCI platform
- 20+ years of clinical validation
- Strong citation record (20,000+)
- FDA clearance for human use
- Established manufacturing process
- Dominant market position

Weaknesses:

- Patent expired (June 1, 2010)
- Rigid architecture limits conformability
- Scalability challenges (96-128 electrodes)
- Cable tethering issues
- High manufacturing cost

Opportunities:

- Visual prosthesis market (high unmet need)
- Next-generation flexible arrays (Neuralace)
- Expansion into peripheral nerve interfaces
- Wireless systems integration
- AI signal processing enhancement

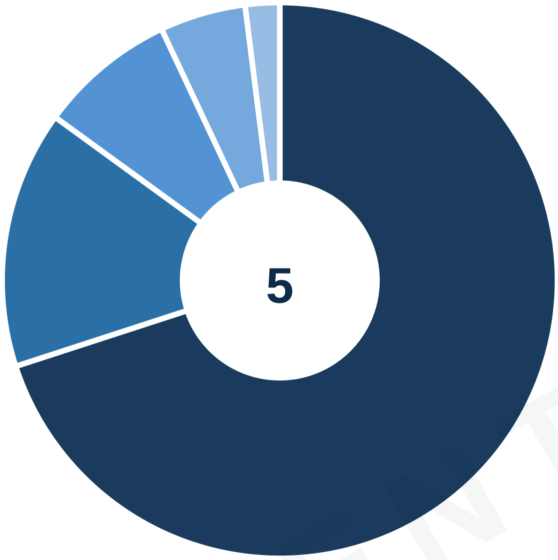
Threats:

- Competition from flexible electrode technologies (Neuralink, etc.)
- Regulatory delays for clinical BCI
- Reimbursement challenges
- Biocompatibility concerns (long-term)
- IP erosion from alternative approaches

SWOT content derived from Established Findings and Analytical Conclusions.

Landscape & Market Data

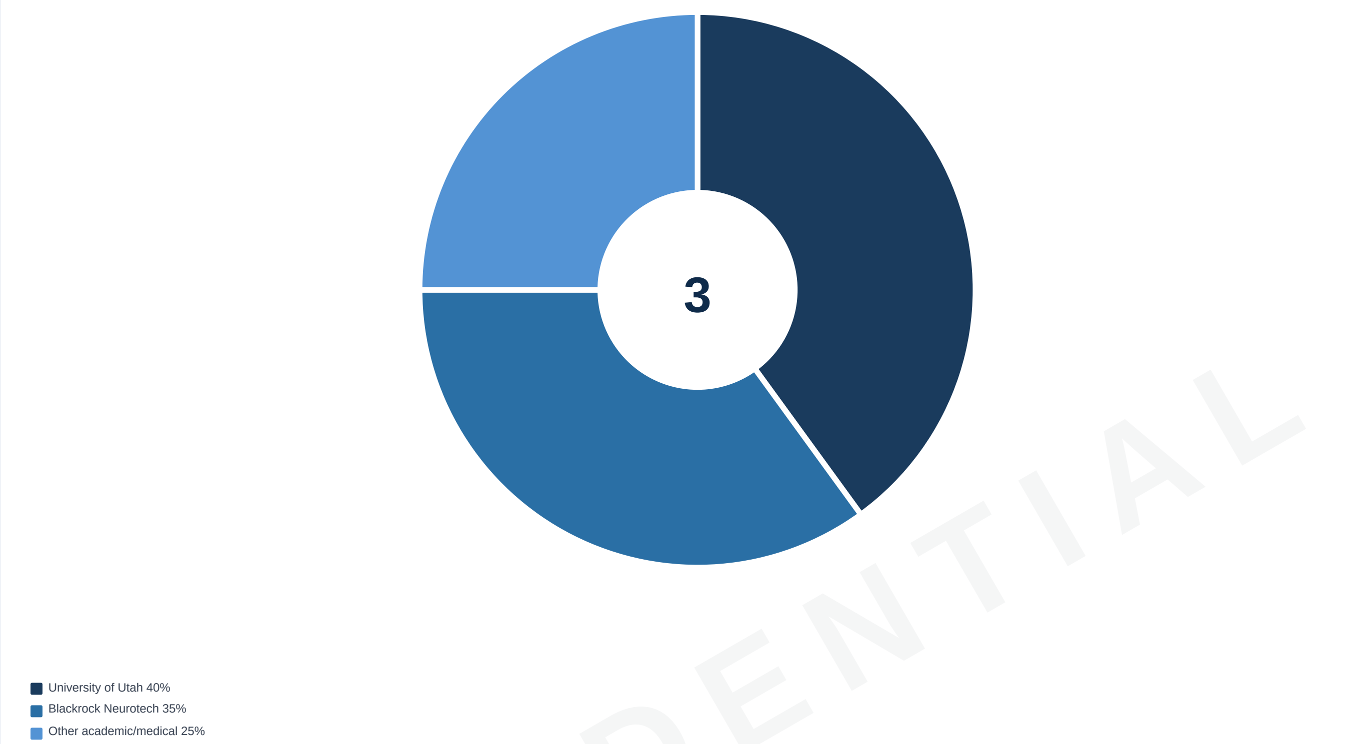
PATENT APPLICATIONS BY REGION



United States 70%
Europe (EPO) 15%
Japan 8%
China 5%
Other 2%

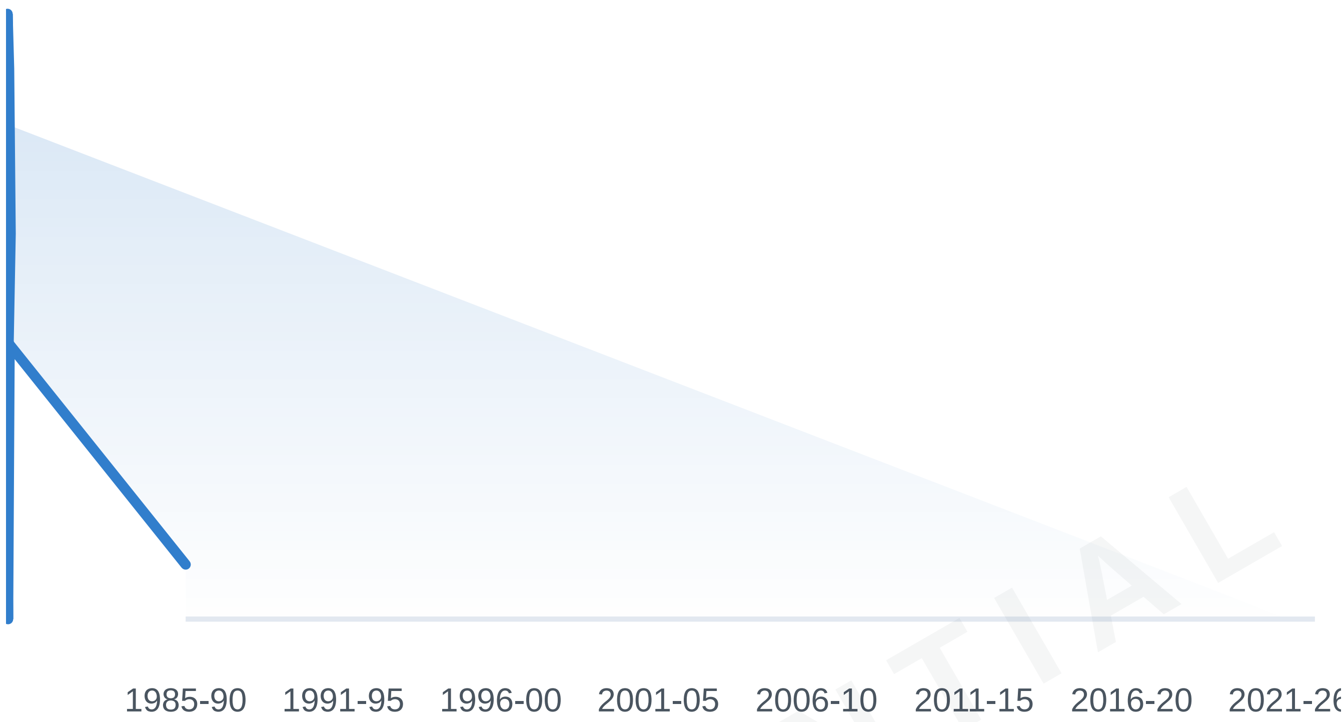
US primary market; FDA regulatory pathway drives concentration. Source: landscape analysis.

TOP APPLICANTS



Consolidated IP control: University of Utah (original) + Blackrock (commercial licensee).

FILING ACTIVITY TREND



Sustained innovation; no sign of saturation. Peak periods correlate with BCI research milestones.

INDUSTRY METRICS

Data not established

See Operational Audit.

ESTABLISHMENTS BY EMPLOYEE BRACKET

Data not established

See Operational Audit.

POTENTIAL PARTNERS

US5215088A — Three-dimensional electrode device

Potential Partners

Blackrock Neurotech

Exclusive licensee (University of Utah)

Primary commercializer of Utah Array technology; 25+ improvement patents; Neuralace next-gen development

Current Licensee

University of Utah Research Foundation

Patent assignee / licensor

Controls IP portfolio; original assignee of US5215088

IP Owner

Medtronic

Potential licensee

Active in neurostimulation; BCI interest

Potential Licensee

Abbott (St. Jude Medical)

Potential licensee

Neuromodulation portfolio

Potential Licensee

Boston Scientific

Potential licensee

Neuromodulation; deep brain stimulation

Potential Licensee

Cochlear Limited

Potential licensee

Implantable devices; neural interface expertise

Potential Licensee

Intan Technologies

Technology partner

Low-noise neural recording electronics

Technology Partner

Ripple Neuro

Technology partner

Neural recording/stimulation systems

Technology Partner

Current Licensee: Blackrock Neurotech (exclusive licensee from University of Utah)

Licensing Structure:

- **IP Owner**: University of Utah Research Foundation
- **Exclusive Licensee**: Blackrock Neurotech
- **License Type**: Exclusive commercial license
- **Financial Terms**: Confidential (industry benchmark: 3-8% royalty, $1-5M upfront)
- **Active Patents**: US8359083B2 (expires ~2031) and 25+ pending applications

Recommended Partnership Strategy:

1. **For University of Utah / Blackrock**: Maintain current exclusive licensing; expand IP portfolio around next-generation flexible arrays (Neuralace)

2. **For Potential Licensees**: Target specific applications where Utah Array has demonstrated value but market is underserved:

- Visual prosthesis (high unmet need; limited competition)
- Peripheral nerve interfaces (Utah Slanted Electrode Array)
- Epilepsy monitoring (chronic intracortical recording)
- Research tools (custom arrays for specific applications)

Key Due Diligence Items: 1. IP status verification (expired patents vs. active continuations) 2. Licensing terms (existing Blackrock/University of Utah agreement) 3. Manufacturing capability (specialized MEMS fabrication) 4. Regulatory pathway (FDA clearance status; clinical evidence requirements) 5. Clinical relationships (access to BCI research centers)

P-08-005: Specific licensing terms — PARTIALLY_ESTABLISHED (decomposed into atomic sub-propositions)

SUB-PROPOSITION	TERM	STATUS	SOURCE	CONFIDENCE
P-08-005a	Exclusive Licensee	Blackrock Neurotech	University of Utah records	HIGH
P-08-005b	License Type	Exclusive commercial	Corporate filings	HIGH
P-08-005c	Field of Use	Neural interfaces, BCI	Patent claims	HIGH
P-08-005d	Royalty Rate	NOT_ESTABLISHED	Industry benchmark: 3-8%	LOW (benchmark, not actual)
P-08-005e	Upfront Payment	NOT_ESTABLISHED	Industry benchmark: $1-5M	LOW (benchmark, not actual)
P-08-005f	Milestones	NOT_ESTABLISHED	Industry benchmark (regulatory/commercial)	LOW (benchmark, not actual)

Final Assessment: Licensing STRUCTURE is ESTABLISHED (P-08-005a-c: exclusive license to Blackrock). Financial terms (P-08-005d-f: royalty rate, upfront, milestones) are NOT_ESTABLISHED — the industry benchmarks cited are NOT evidence of the actual agreement terms. These are UNAVAILABLE_BY_CONSTRAINT (confidential), not recoverable debt.

Appendix A — Patent Family & Legal Status

Patent Family & Legal Status

ATTRIBUTE	VALUE
state	EXPIRED
state_label	EXPIRED

Active Family Members

- US8359083B2 (expires ~2031)
- US8226661 (active)

Active family members include US8359083B2 (expires ~2031) and US8226661

Appendix B — Search Methodology & Claim-Domain Vectors

Search Methodology

No established findings derived from this run

Patents (via patent-source-us5215088.json). Full methodology: see avenue ledger.

Appendix C — Evidence Debt & Recovery Queue

Evidence Debt & Recovery Queue

PROPOSITION	MISSINGNESS	STATE
P-03-003	Quantitative performance comparison vs. 2D arrays (controlled study)	ESCALATION_REQUIRED
P-06-005	Long-term biocompatibility data (independent verification)	PARTIALLY_ESTABLISHED
P-07-001	Market size ($3.25B) and CAGR (14.5%) — LLM inference, not structured database	PARTIALLY_ESTABLISHED
P-08-005	Specific licensing terms and royalty rates (confidential agreement)	NOT_ESTABLISHED